PCBC H Name _____ _____ /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

PRACTICE D · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

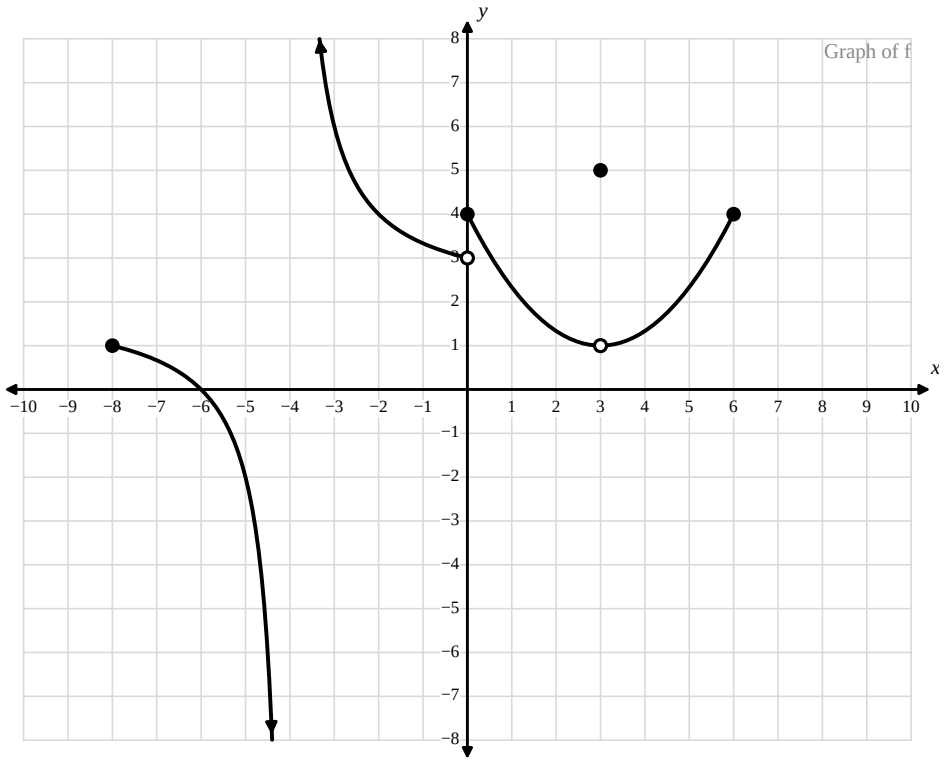
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Signature _____

SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)

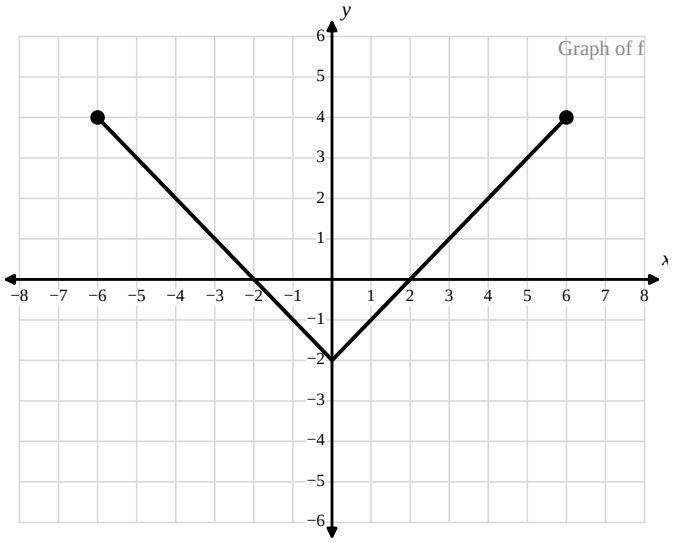


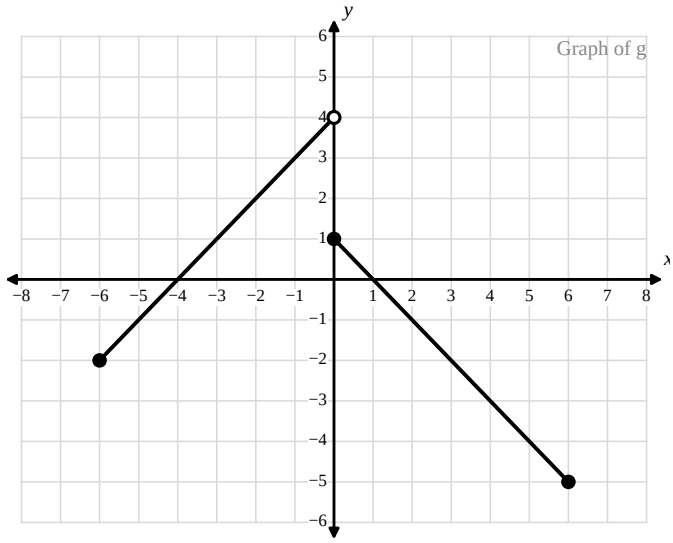
1. $\lim_{x\to 3} f(x) =$	**2.** $\lim_{x\to -4^-} f(x) =$	**3.** $\lim_{x\to 0} f(x) =$

(2 points)

4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = 4$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim\limits_{x \to 0^-} \dfrac{f(x)}{g(x)}$

(2 points)

6. Use the table to estimate $\lim\limits_{x \to 0} k(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	−0.1	−0.01	−0.001	→ 0 ←	0.001	0.01	0.1
$k(x)$	0.951626	0.995017	0.999500		1.000500	1.005017	1.051709

$\lim\limits_{x \to 0^-} k(x) \approx$ _____ $\lim\limits_{x \to 0^+} k(x) \approx$ _____ $\lim\limits_{x \to 0} k(x) \approx$ _____

Evaluate each limit. (2 points each)

7. $\lim\limits_{x\to\frac{\pi}{3}} (6\cos x - \tan x)$	8. $\lim\limits_{x\to -2} \dfrac{x^3 - 4x}{x^2 + 2x}$
9. $\lim\limits_{h\to 0} \dfrac{\sqrt{25+h} - 5}{h}$	10. $\lim\limits_{x\to 1} \dfrac{\frac{2}{x+1} - 1}{x - 1}$
11. For $f(x) = 3x^2 + x$, find $\lim\limits_{h\to 0} \dfrac{f(x+h) - f(x)}{h}$	

Answer Key · Practice D

Each step: the work on the left, the reason on the right. "lim" stays on every line until the line where you substitute.

1. $\boxed{1}$

Both sides head to the open circle at $(3, 1)$.	The dot $(3, 5)$ is $f(3)$ and does not matter.

2. $\boxed{-\infty}$

Just left of $x = -4$ the curve falls without bound.	Left side only. (From the right it rises to $+\infty$.)

3. $\boxed{\text{DNE}}$

$\lim_{x \to 0^-} f(x) = 3, \ \lim_{x \to 0^+} f(x) = 4$	Left: open circle $(0, 3)$. Right: the curve starts at the filled dot $(0, 4)$.
$3 \neq 4$, so $\lim_{x \to 0} f(x)$ does not exist.	Sides disagree.

4. $\boxed{a = -2}$

Height 4 is reached from both sides only at $x = -2$.	Two-sided approach.
Not $x = 0$ (right side only; the left heads to 3) and not $x = 6$ (the graph ends there).	One-sided spots and endpoints do not count.

5. $\boxed{-\frac{1}{2}}$

$\lim_{x \to 0^-} \dfrac{f(x)}{g(x)} = \dfrac{\lim_{x \to 0^-} f(x)}{\lim_{x \to 0^-} g(x)}$	Limit of a quotient; allowed because the bottom limit is 4, not 0.
$= \dfrac{-2}{4} = -\dfrac{1}{2}$	$f \to -2$; from the left g heads to the open circle $(0, 4)$. Trap: $g(0) = 1$ would give -2.

6. $\boxed{\approx 1}$

Left: 0.951626, 0.995017, 0.999500	Heading to 1.
Right: 1.051709, 1.005017, 1.000500	Heading to 1.
$\lim_{x \to 0} k(x) \approx 1$	Both sides agree. An estimate, not a proof.

7. $\boxed{3 - \sqrt{3}}$

$\lim_{x \to \frac{\pi}{3}} (6 \cos x - \tan x) = 6 \cos \dfrac{\pi}{3} - \tan \dfrac{\pi}{3}$	$\pi/3$ is in the domain: substitute.
$= 6\left(\dfrac{1}{2}\right) - \sqrt{3} = 3 - \sqrt{3}$	Unit circle.

8. $\boxed{-4}$

Substitution produces 0/0: indeterminate.	Factor.
$\lim_{x \to -2} \dfrac{x^3 - 4x}{x^2 + 2x} = \lim_{x \to -2} \dfrac{x(x-2)(x+2)}{x(x+2)}$	GCF x, then a difference of squares on top.
$= \lim_{x \to -2} (x - 2) = -2 - 2 = -4$	Cancel x and $x + 2$; substitute.

9. $\boxed{\frac{1}{10}}$

Substituting $h = 0$ produces 0/0.	Multiply by the conjugate $\sqrt{25+h}+5$.
$\lim\limits_{h\to 0} \dfrac{\sqrt{25+h}-5}{h} = \lim\limits_{h\to 0} \dfrac{(25+h)-25}{h\left(\sqrt{25+h}+5\right)}$	Top is a^2-b^2; bottom stays factored.
$= \lim\limits_{h\to 0} \dfrac{1}{\sqrt{25+h}+5} = \dfrac{1}{5+5} = \dfrac{1}{10}$	Cancel h; substitute 0 for h.

10. $\boxed{-\frac{1}{2}}$

Substitution produces 0/0: indeterminate.	Multiply top and bottom by the LCD $x+1$.
$\lim\limits_{x\to 1} \dfrac{\frac{2}{x+1}-1}{x-1} = \lim\limits_{x\to 1} \dfrac{2-(x+1)}{(x+1)(x-1)}$	$(x+1)\cdot\frac{2}{x+1} = 2$, $(x+1)\cdot 1 = x+1$.
$= \lim\limits_{x\to 1} \dfrac{-(x-1)}{(x+1)(x-1)} = \lim\limits_{x\to 1} \dfrac{-1}{x+1}$	$2-x-1 = 1-x = -(x-1)$; cancel.
$= \dfrac{-1}{1+1} = -\dfrac{1}{2}$	Substitute.

11. $\boxed{6x+1}$

$\lim\limits_{h\to 0} \dfrac{\left[3(x+h)^2+(x+h)\right]-\left(3x^2+x\right)}{h}$	Build $f(x+h)$ with brackets.
$= \lim\limits_{h\to 0} \dfrac{3x^2+6xh+3h^2+x+h-3x^2-x}{h}$	Expand; distribute the negative.
$= \lim\limits_{h\to 0} \dfrac{h(6x+3h+1)}{h} = \lim\limits_{h\to 0}(6x+3h+1)$	Factor h; cancel.
$= 6x+0+1 = 6x+1$	Substitute 0 for h.